Percent Yield and Stoichiometry

- **Percent Yield** describes how much product was actually made in the lab versus the amount that theoretically could be made.

$$\frac{\text{Actual Yield}}{\text{Theoretical Yield}} \cdot 100 = \text{Percent Yield}$$

- Percent yield tells you how close you were to the 100% mark.

- Reactions do not always work perfectly. Experimental error (spills, contamination) often means that the amount of product made in the lab does not match the ideal amount that could have been made.

- **Theoretical Yield** = The maximum amount of product that could be formed from given amounts of reactants.

- **Actual Yield** = The amount of product actually formed or recovered when the reaction is carried out in the laboratory.

What might reduce yield?

Brainstorm some ideas, thinking of experiments you have done or even baking and cooking?

Why did you not get the maximum amount of product from the reactant you put in?

Check your ideas in comparison to the next slide

Next slide, type in student ideas in yellow box.

Move to see teacher list once done

Factors affecting yield?

- Side reactions with unpredicted chemicals
- Undesired products
- Insufficient reactants
- Loss of reactants or products
- Errors in process (eg overheating)
- Not well mixed
- Loss during transfer

Example:

- A chemist was supposed to produce 75.0 g of aspirin. However, he squandered some of the reactants for his own personal use (he was later fired), and so only actually made 50.0 g. What was his percent yield?

$$\frac{\mathbf{50.0\ g}}{\mathbf{75.0\ g}} \times \mathbf{100} \quad = \mathbf{66.7\ \%}$$

More Examples!

- When I was a sophomore in college, we had a lab where we were supposed to isolate caffeine from tea leaves. Most of my caffeine was washed down the drain in a freak accident. Although I should have had 5.0 g of caffeine, I only ended up with 0.040 g of caffeine and a bad grade on the lab. What was my percent yield?

$$\frac{0.040\ \text{g}}{5.0\ \text{g}} \times 100 = 0.80\ \%$$

2. A very sloppy student did not wait until his NaCl was dry before he weighed it. As a result, his product weighed 2.25 g when it should have been 1.75 g. What was his percent yield?

$$\frac{2.25\ \text{g}}{1.75\ \text{g}} \times 100 = 129\%$$

3. What is the theoretical yield if 5.50 grams of hydrogen react with excess nitrogen to form ammonia?

$$\mathbf{3}\ H_2 + N_2 \rightarrow \mathbf{2}\ NH_3$$

Do YOUR ARROW JOURNEY

= 30.93 grams NH_3 = Theoretical Yield!!!!

Only 20.4 grams of ammonia is actually produced in the lab. What is the percent yield?

$$\frac{\mathbf{20.4\ g}}{\mathbf{30.93\ g}} \ \mathbf{x\ 100} = 65.96\ \%\ \text{yield}$$